

The Global Set of Attainable Unitary Factors

A Literature-Implied Complete Characterization

Status note for arXiv:2605.28885

12 August 2026

Status: literature-implied answer. This is an agent-identified consequence of a 1986 matrix-factorization theorem, not an original result and not an answer knowingly directed at the 2026 source paper.

1 Source question

Vuong's arXiv:2605.28885 proves a fixed-pair criterion for the unitary factors of generalized fidelity [1]. In the Discussion (PDF p. 27), the paper identifies

$$\mathcal{G}_d := \bigcup_{P, Q \in \mathbb{P}_d} \{W \in U(d) : P^{-1/2}Q^{1/2}W \text{ is similar to a positive-definite matrix}\} \quad (1)$$

as the natural global object to characterize more explicitly.

$$F_R(P, Q) = F^H(P, Q),$$

showing that the sufficient polar condition captures only one distinguished stratum of the full solution set. The polar slice corresponds exactly to the unitary stratum $W = I$, while Example 5.7 shows that nonpolar Holevo bases do occur.

The classification of unitary factors also settles the special-unitary reverse-inclusion question in the negative in every dimension $d \geq 2$. Although every generalized-fidelity unitary factor lies in $SU(d)$, not every element of $SU(d)$ can arise as such a unitary factor when $d \geq 2$. In fact, for every $d \geq 2$, every nontrivial central special unitary ωI , with $\omega^d = 1$ and $\omega \neq 1$, is excluded by the fixed-pair criterion

$$W \text{ can arise as a unitary factor for } (P, Q) \iff P^{-1/2}Q^{1/2}W \sim \mathbb{P}_d.$$

Thus Open Problem 7 of [1] has a negative answer in every dimension $d \geq 2$.

The main questions that remain open are therefore of a different nature. On the polar-fidelity side, the principal unresolved issue is full data-processing monotonicity beyond commutative interfaces, together with related convexity questions for generalized Bures distance. On the generalized-fidelity side, the global set of arising unitary factors

$$\bigcup_{P, Q \in \mathbb{P}_d} \{W \in U(d) : P^{-1/2}Q^{1/2}W \sim \mathbb{P}_d\}$$

now appears as the natural object to characterize more explicitly. It would also be interesting to determine whether semidefinite-program formulations are available for the real part of

Figure 1: The stated global object in arXiv:2605.28885, p. 27.

2 Identification with a classical factorization class

We use the standard fact, also implicit in the source paper, that a complex matrix is similar to a positive-definite matrix if and only if it is a product of two positive-definite matrices.

Attainable implies four factors. If $W \in \mathcal{G}_d$, put $M = P^{-1/2}Q^{1/2}$ for a witnessing pair. Both M and MW are products of two positive-definite matrices. Since the inverse of a positive-definite matrix is positive definite,

$$W = M^{-1}(MW)$$

is a product of four positive-definite matrices.

Four factors imply attainable. Conversely, suppose

$$W = H_1 H_2 H_3 H_4, \quad H_j > 0.$$

Set

$$P = H_2^2, \quad Q = H_1^{-2}.$$

Then

$$M = P^{-1/2}Q^{1/2} = H_2^{-1}H_1^{-1}, \quad MW = H_3H_4.$$

The latter is a product of two positive-definite matrices and hence is similar to a positive-definite matrix. The fixed-pair criterion of [1, Theorem 5.9] therefore makes W attainable. We have proved

$$W \in \mathcal{G}_d \iff W \text{ is a product of four positive-definite matrices.} \quad (2)$$

3 Sourour–Ballantine criterion and the answer

Sourour’s Theorem 2, attributed there to Ballantine, states that a real or complex matrix A is a product of four positive-definite matrices if and only if

$$\det A > 0 \quad \text{and} \quad A \neq \alpha I \text{ for every scalar } \alpha \text{ that is not positive.} \quad (3)$$

Here $\det A > 0$ means a positive real determinant [2].

For a unitary W , the first condition in (3) is equivalent to $\det W = 1$, namely $W \in SU(d)$. The scalar elements of $SU(d)$ are ωI with $\omega^d = 1$. Among them, only I has a positive scalar. Combining (2) and (3) gives the complete characterization

$$\boxed{\mathcal{G}_d = SU(d) \setminus \{\omega I : \omega^d = 1, \omega \neq 1\}.} \quad (4)$$

This sharpens Corollary 5.15 of the source: the nonidentity central special unitaries are not merely examples of excluded factors; they are the *only* excluded elements of $SU(d)$.

4 Provenance and remaining scope

The supporting theorem predates the source by forty years and does not mention quantum fidelity. The source paper does not cite Ballantine or Sourour. Accordingly, (4) is classified as a *literature-implied answer*: the mathematical input is known, while the identification with (1) is the bridge recorded here.

2. PRODUCTS OF POSITIVE DEFINITE MATRICES

In [1] and [2], Ballantine proved that every real or complex $n \times n$ matrix A with positive determinant is a product of five positive-definite matrices and, unless A is a nonpositive scalar, it is a product of four positive-definite matrices, (see also Taussky [9]). We give very short proofs of these facts using our Theorem 1. (Ballantine also characterized products of three positive definite matrices.)

THEOREM 2 (Ballantine). Let A be a real or complex $n \times n$ matrix. Then

- (a) A is a product of four positive-definite matrices if and only if $\det A > 0$ and A is not a scalar αI where α is not positive.
- (b) A is a product of five positive-definite matrices if and only if $\det A > 0$.

Figure 2: Sourour [2, Theorem 2], reproducing Ballantine’s four-positive-factor criterion.

The bounded search checked the run’s cheap indexes; the source’s references; exact searches for products of four positive-definite matrices; Ballantine’s 1968/1970 papers; Sourour’s primary 1986 paper; and arXiv:2507.12560 (2025), which surveys constructive positive-factor decompositions. No evidence was found that the 2026 source authors had already made this identification.

This note resolves only the explicit global-unitary-factor object. Full data-processing monotonicity, related convexity questions, and the proposed semidefinite-program formulations remain open.

References

- [1] T. D. Vuong, *Polar Fidelities, Holevo Bases, and Unitary Factors of Generalized Fidelity*, arXiv:2605.28885, 2026.
- [2] A. R. Sourour, *A Factorization Theorem for Matrices*, Linear and Multilinear Algebra **19**(2) (1986), 141–147, doi:10.1080/03081088608817711.